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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 10

Team No.: 20

Project Title: Linux System Information and Resource Monitoring Tool

Team Members

Roll Number	Student Name	Signature
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1. Abstract

The Linux System Information and Resource Monitoring Tool is a Linux-based system programming project designed to collect, display, and monitor important system information and resource utilization. The tool provides users with a simple interface to observe CPU usage, memory utilization, disk usage, running processes, system uptime, and other essential system details. The project demonstrates practical Operating Systems and Systems Programming concepts by interacting with Linux system resources through system calls, POSIX APIs, and the /proc filesystem. The proposed solution will be implemented using C on Ubuntu and will periodically collect system information and present it in a clear and understandable format. Process-related information can be obtained from the Linux process management interfaces, while CPU and memory statistics can be retrieved from system-provided information. The tool will help users understand how operating systems manage hardware resources and processes. The expected outcome is a lightweight monitoring utility capable of providing real-time or periodic information about system performance and resource consumption. This project provides practical exposure to Linux system programming, process management, file operations, system information retrieval, and resource monitoring.

2. **Problem Statement**

Monitoring system resources is important for understanding the performance and health of a Linux system. However, beginners may find it difficult to understand how CPU, memory, disk space, processes, and other system resources are managed and accessed by the operating system. Existing monitoring utilities may also provide large amounts of information that can be difficult for students to interpret. The proposed Linux System Information and Resource Monitoring Tool aims to develop a simple Linux-based application that collects and displays essential system information and resource usage in an organized manner. The project focuses on applying Operating Systems and Systems Programming concepts to access system information and monitor resources using Linux mechanisms and APIs.

3. Objectives

The project aims to collect and display essential system information such as CPU details, memory information, system uptime, and disk usage. It also focuses on monitoring important system resources, including CPU and memory utilization, at regular intervals. The tool will retrieve and display information about running processes to help users understand process management in Linux. Furthermore, the project will implement the monitoring tool using Linux system programming concepts and APIs, providing practical understanding of operating system resource management.

4. Proposed Methodology

The project will be designed and implemented as a Linux-based monitoring application using C/C++. The application will collect system information from Linux system interfaces and present the information in a structured format.The major functionality will include collecting CPU statistics, memory information, disk usage, system uptime, and running-process details. Linux system interfaces such as the /proc filesystem will be used to obtain relevant system and process information. Appropriate POSIX/Linux APIs and system calls will be incorporated where required.The application will periodically retrieve resource statistics, process the collected information, and display the results to the user. The basic flow will be:

Start → Collect System Information → Read Resource Statistics → Retrieve Process Information → Process Data → Display Results → Periodically Refresh → Exit

The implementation will be tested on a Linux/Ubuntu environment to verify that the monitoring information is collected and displayed correctly.

5. Operating Systems Concepts / Linux APIs Used

The project uses several important Operating System concepts and Linux APIs to monitor and retrieve system-level information. Process Management is used to retrieve and monitor information about currently running processes, including their resource usage and status. The /proc Filesystem provides access to CPU, memory, uptime, and process-related information maintained by the Linux kernel. File Operations are used to read system information exposed through Linux system files. POSIX APIs are used to interact with Linux operating-system resources and provide system-level functionality. Process Information is used to display details about running processes and their resource consumption. System Calls demonstrate how the application interacts directly with the Linux operating system to perform system-level operations. CPU Resource Management is used to monitor CPU utilization and understand how processor resources are being used. Memory Management is used to retrieve and display information about memory usage, available memory, and overall memory resources. Disk Management is used to monitor the available and used disk space of the system. Finally, System Uptime is used to retrieve and display the amount of time the Linux system has been running since it was last started.

6. Individual Contribution

The project involved designing and developing the complete Linux system monitoring application. Process Management was implemented to retrieve and monitor currently running processes along with their resource usage and status. The /proc Filesystem was used to obtain CPU, memory, uptime, and process-related information maintained by the Linux kernel. File Operations were implemented to read system information from Linux system files, while POSIX APIs were used to interact with operating-system resources. Process Information features were developed to display details about running processes and their resource consumption. System Calls were used to demonstrate interaction between the application and the Linux operating system. CPU Resource Management and Memory Management were implemented to monitor CPU utilization, memory usage, and memory availability. Disk Management was used to monitor available and used disk space, while System Uptime functionality was implemented to retrieve and display the time for which the Linux system has been running. Coding, testing, debugging, error handling, documentation, and presentation of the complete project were also carried out.

7. Tools / Platforms / Software Used

The project uses Ubuntu as the development and execution environment, while C Programming is used for the core project implementation. GCC is used for compiling the C programs, and Linux/POSIX APIs are used for process management and system programming. VS Code is used for source code development, while Git and GitHub are used for version control and maintaining the project repository. Makefile is used for build and compilation management.

8. Expected Outcome

The project is expected to provide a functional Linux System Information and Resource Monitoring Tool through a command-line interface. The system will allow users to retrieve and display important system information such as CPU utilization, memory usage, disk usage, system uptime, and other system-related details. It will use the /proc filesystem, Linux system files, POSIX APIs, and system calls to obtain real-time system information. The project will demonstrate practical understanding of Linux system programming, resource management, file operations, and operating system concepts, resulting in a simple and efficient tool for monitoring the overall health and resource utilization of a Linux system.

9. GitHub Repository

GitHub Repository Link: https://github.com/gouthamvikasparimi-eng/Operating-Systems-and-Systems-Programming

Repository Created: Yes

Faculty Name and Signature

Faculty Name: _____

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____